🌡️ Climate Analysis Spain — Temperature Trends (1950–2024)

Are Spanish cities getting warmer? This project analyses historical daily temperature data for 5 major Spanish cities over 7 decades, using Python and open climate data.

📊 Key Findings

- All 5 cities show a **clear warming trend** between 1950 and 2024.
- Average warming across cities: ~**+1.5°C** over 70 years.
- The warming trend accelerates noticeably from the **1980s onwards**.
- Interior cities (Madrid, Seville) show stronger warming than coastal ones (Bilbao, Barcelona).

(Exact values are generated dynamically from the data when you run the script.)

📁 Project Structure

```
climate-analysis-spain/
│
├── climate_analysis.py         # Main analysis script
├── climate_spain_analysis.png # Output chart (generated on run)
└── README.md
```

🛠️ Tools & Libraries

Tool	Purpose
Python 3.x	Main language
Pandas	Data loading, cleaning and aggregation
Matplotlib	Visualisation
Seaborn	Chart styling
NumPy	Linear regression (trend lines)
Open-Meteo API	Free historical climate data (no API key needed)

🚀 How to Run

1. Clone this repository:

```bash
git clone https://github.com/YOUR_USERNAME/climate-analysis-spain.git
cd climate-analysis-spain
```

2. Install dependencies:

```bash
pip install pandas matplotlib seaborn requests numpy
```

3. Run the analysis:

```bash
python climate_analysis.py
```

The script will:

- Download historical temperature data automatically (no manual download needed)
- Print a statistical summary in the terminal
- Save a multi-panel chart as `climate_spain_analysis.png`

📈 Visualisations

The output chart includes three panels:

1. **Annual mean temperature (10-year moving average)** — shows the long-term warming trend for each city
2. **Total warming per city** — compares warming between the 1950s and 2020s
3. **Monthly temperature profile** — seasonal patterns across the full series

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📡 Data Source

Open-Meteo Historical Weather API — free, no registration required, based on ERA5 reanalysis data from the Copernicus Climate Change Service.

👤 Author

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📝 Notes

This project was developed as part of a personal data analysis portfolio, combining a background in geophysics with practical Python data analysis skills. The methodology is intentionally straightforward to focus on clarity of insight over complexity of implementation.